

Nutrition Assistant Project

Phase 2: Requirement Analysis

Introduction

The Requirement Analysis phase identifies the functional and non-functional requirements of the Nutrition Assistant application. This phase ensures that the development team clearly understands the expected behavior, performance, security, and usability of the system before implementation begins.

Objectives

Gather user requirements. Define system functionalities. Specify quality attributes. Reduce ambiguity during development. Provide a foundation for system design and testing.

Functional Requirements

User Authentication: Users can register, log in, and securely log out. Profile Management: Users can update personal information such as age, weight, height, and dietary preferences. Food Database Search: Search foods and retrieve nutritional information. Calorie & Macro Calculation: Automatically calculate calories, protein, carbohydrates, and fats. Meal Logging: Record breakfast, lunch, dinner, and snacks. Dashboard: Display daily nutritional intake, charts, and progress. Goal Tracking: Compare consumed calories with target goals. Notifications: Meal reminders and water intake alerts.

Non-Functional Requirements

Performance: API responses should be below 200 milliseconds under normal conditions. **Scalability:** Support thousands of concurrent users. **Reliability:** Ensure high availability with minimal downtime. **Responsiveness:** Responsive UI for mobile, tablet, and desktop. **Security:** Encrypt passwords and sensitive user data using secure algorithms. **Usability:** Provide a simple, intuitive, and accessible interface. **Maintainability:** Modular architecture for future enhancements.

User Stories

As a user, I want to create an account so my nutrition data is stored securely. As a user, I want to search for an apple so I can log its calories. As a user, I want to record my daily meals to monitor my eating habits. As a user, I want to view charts and summaries to understand my nutrition progress. As a user, I want to receive reminders so I never miss meals or water intake.

Expected Deliverables

The outcome of this phase is a complete Software Requirement Specification (SRS) containing all functional requirements, non-functional requirements, user stories, system constraints, and acceptance criteria. This document serves as the reference for the design and implementation phases.

Conclusion

Requirement Analysis ensures that the Nutrition Assistant project is developed according to user expectations and quality standards. A well-defined requirement document minimizes development risks, improves communication among stakeholders, and increases the likelihood of delivering a successful application.